Cluster Health Monitoring

> **Series:** K8S | **Notebook:** 4 of 12 | **Created:** January 2026 | **Last Updated:** 02/05/2026

Deep-Dive into Kubernetes Cluster Metrics

Cluster health monitoring provides visibility into the infrastructure layer of Kubernetes: nodes, control plane, and cluster-wide resources. This notebook covers key metrics, thresholds, and DQL queries for proactive cluster management.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Kubernetes monitoring
DynaKube	ActiveGate with `kubernetes-monitoring` capability
Permissions	`metrics.read`, `entities.read`, `logs.read`
Data	At least 24 hours of cluster data

1. Cluster Health Overview

Key Health Indicators

Category	Metrics	Healthy State
Node Status	Ready/NotReady	All nodes Ready
Pod Scheduling	Pending pods	No long-pending pods
Resource Pressure	CPU/Memory pressure	No pressure conditions
Disk Pressure	Disk space, inode usage	>15% available
Network	CNI health, DNS latency	<100ms DNS resolution

Dynatrace Kubernetes Dashboard

The built-in Kubernetes dashboard provides:
- Cluster overview with node status
- Namespace resource usage
- Workload health summary
- Recent events and problems

Navigate to: **Infrastructure > Kubernetes**

```
// List all monitored Kubernetes clusters
fetch dt.entity.kubernetes_cluster
| fields entity.name, tags
| sort entity.name asc
```

2. Node Monitoring

Node Status and Conditions

Condition	Description	Alert When
Ready	Node can accept pods	False for >5 min
MemoryPressure	Low memory	True
DiskPressure	Low disk space	True
PIDPressure	Too many processes	True
NetworkUnavailable	Network not configured	True

```
// List all Kubernetes nodes
fetch dt.entity.kubernetes_node
| fields entity.name, tags
| sort entity.name asc
```

```
// Node CPU utilization over time (top 10 by name)
timeseries nodeCpu = avg(dt.kubernetes.node.cpu_usage), from:-1h, by:
{dt.entity.kubernetes_node}
| limit 10
```

```
// Node memory utilization - average over time period
timeseries avgMemory = avg(dt.kubernetes.node.memory_usage), from:-1h, by:
{dt.entity.kubernetes_node}
| fieldsAdd avgMemoryValue = arrayAvg(avgMemory)
| filter avgMemoryValue &gt; 80
| sort avgMemoryValue desc
```

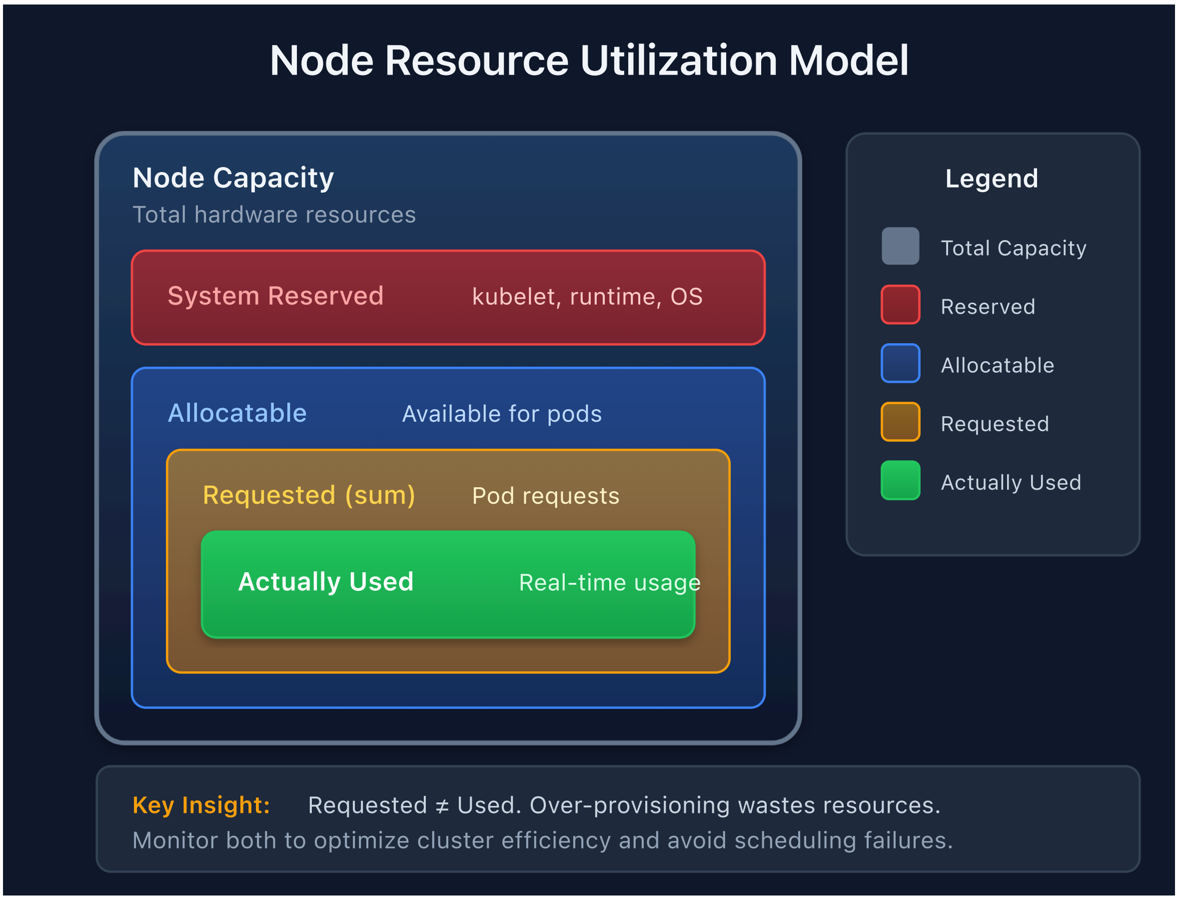
```
// Node filesystem usage - disk pressure detection
timeseries avgDiskUsage = avg(dt.host.disk.used.percent), from:-1h, by:
{dt.entity.host}
| fieldsAdd avgDiskUsageValue = arrayAvg(avgDiskUsage)
| filter avgDiskUsageValue &gt; 80
| sort avgDiskUsageValue desc
```

3. Resource Capacity Planning

Capacity Metrics

Metric	Description	Use Case
Allocatable	Resources available for pods	Scheduling decisions
Requested	Sum of pod requests	Capacity planning
Used	Actual consumption	Right-sizing
Limits	Maximum allowed	Burst capacity

Utilization vs. Allocation




```
// CPU requests by namespace (average over time period)
timeseries avgCpuRequests = avg(dt.kubernetes.workload.requests_cpu),
from:-1h, by:{k8s.namespace.name}
| sort avgCpuRequests desc
| limit 15
```

```
// Memory requests by namespace (average over time period)
timeseries avgMemRequests = avg(dt.kubernetes.workload.requests_memory),
from:-1h, by:{k8s.namespace.name}
| sort avgMemRequests desc
| limit 15
```

```
// Find over-provisioned workloads (low CPU usage)
timeseries avgCpuUsageMillicores = avg(dt.kubernetes.container.cpu_usage),
from:-1h, by:{dt.entity.cloud_application}
| fieldsAdd avgCpuUsageMillicoresValue = arrayAvg(avgCpuUsageMillicores)
| sort avgCpuUsageMillicoresValue asc
| limit 20
```

4. Control Plane Health

Control Plane Components

Component	Function	Key Metrics
API Server	REST API for K8s	Request latency, error rate
etcd	Distributed KV store	Disk sync latency, leader elections
Scheduler	Pod placement	Scheduling latency, failures
Controller Manager	Reconciliation loops	Queue depth, sync latency

Managed Kubernetes Note

For managed Kubernetes (EKS, AKS, GKE), control plane metrics are limited. Focus on:
- API server response times (client-side)
- Kubernetes events for scheduling issues
- Cloud provider metrics for control plane health

```
// API server events and errors
fetch logs, from:-1h
| filter matchesPhrase(content, "kube-apiserver") or matchesPhrase(content,
"api-server")
| filter matchesPhrase(content, "error") or matchesPhrase(content, "failed")
| fields timestamp, content
| sort timestamp desc
| limit 20
```

5. Cluster-Wide Events

Event Types to Monitor

Event Type	Reason	Action
Warning	FailedScheduling	Check resource constraints
Warning	FailedMount	Check PV/PVC configuration
Warning	OOMKilled	Increase memory limits
Warning	Evicted	Node under pressure
Normal	Pulling/Pulled	Image operations
Normal	Scheduled	Pod placement

```
// Kubernetes warning events
fetch logs, from:-1h
| filter matchesPhrase(content, "Warning") and (matchesPhrase(log.source,
"kubernetes") or matchesPhrase(log.source, "k8s"))
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// Failed scheduling events
fetch logs, from:-1h
| filter matchesPhrase(content, "FailedScheduling") or matchesPhrase(content,
"Insufficient")
| fields timestamp, content
| sort timestamp desc
| limit 30
```

```
// OOMKilled events - memory issues
fetch logs, from:-1h
| filter matchesPhrase(content, "OOMKilled") or matchesPhrase(content, "Out
of memory")
| fields timestamp, content
| sort timestamp desc
| limit 30
```

```
// Event summary by type
fetch logs, from: now() - 24h
| filter matchesPhrase(log.source, "kubernetes") or matchesPhrase(log.source,
"k8s")
| parse content, "LD:eventType ' ' LD"
| summarize count = count(), by:{eventType}
| sort count desc
| limit 20
```

6. Cost Optimization Queries

Resource Efficiency Analysis

Metric	Target	Action If Not Met
CPU Utilization	>40% avg	Reduce requests
Memory Utilization	>50% avg	Reduce requests
Node Utilization	>60%	Scale down nodes
Idle Pods	0	Review necessity

```
// Find workloads with very low CPU utilization (candidates for right-sizing)
timeseries avgCpuUsageMillicores = avg(dt.kubernetes.container.cpu_usage),
from:-1h, by:{dt.entity.cloud_application}
| fieldsAdd avgCpuUsageMillicoresValue = arrayAvg(avgCpuUsageMillicores)
| sort avgCpuUsageMillicoresValue asc
| limit 25
```

```
// Memory usage efficiency by workload (low usage = over-provisioned)
timeseries avgMemUsageBytes =
avg(dt.kubernetes.container.memory_working_set), from:-1h, by:
{dt.entity.cloud_application}
| fieldsAdd avgMemUsageBytesValue = arrayAvg(avgMemUsageBytes)
| sort avgMemUsageBytesValue asc
| limit 25
```

7. Alerting Strategies

Recommended Alerts

Alert	Condition	Severity
Node NotReady	Node condition != Ready for 5 min	Critical
High Node CPU	CPU > 85% for 15 min	Warning
High Node Memory	Memory > 90% for 10 min	Critical
Disk Pressure	Disk > 85%	Warning
Pod Scheduling Failed	FailedScheduling events	Warning
OOM Kills	OOMKilled events	Warning

Alert Configuration in Dynatrace

Navigate to: **Settings > Anomaly detection > Kubernetes**

Configure:
- Node availability alerts
- Resource saturation thresholds
- Workload health anomalies

Custom Metric Events

For advanced alerting, use custom metric events with DQL-derived thresholds.

Next Steps

With cluster health monitoring in place, proceed to:

Next Notebook	Topic
K8S-05: Workload Monitoring	Application-level observability
K8S-06: Namespace Organization	Boundaries and access control
K8S-07: Events and Logs	Log ingestion and analysis

Summary

In this notebook, you learned:

- Cluster health overview and key indicators
- Node monitoring for CPU, memory, and disk
- Resource capacity planning with requests vs. usage analysis
- Control plane health considerations
- Cluster-wide event monitoring and analysis
- Cost optimization queries for right-sizing
- Alerting strategies for proactive cluster management

References

- Kubernetes Cluster Monitoring
- Kubernetes Metrics
- Kubernetes Anomaly Detection
